VISVESVARAYA TECHNOLOGICAL UNIVERSITY

BELAGAVI-590014

A Mini-Project Report **submitted by**

Date of submission: 23/05/2026

Team members details			
SN	**Name**	**USN**	**Section**
1	**G Sindhu**	**1CR24EC071**	**B**
2	**Kavya r s**	**1CR24EC097**	**B**
3	**Sandhya L**	**1CR25EC410**	**B**

Under the guidance of

SN	**Project guide name**	**Designation**	**Department**
1	**Prof.Geethanjali Pawanekumar**	**Professor**	**ECE**

CMR INSTITUTE OF TECHNOLOGY

132 AECS Layout, ITPL Main Road, Kundalahalli Bangalore 560037, India 2026-27

ABSTRACT

In the present era of rapid technological advancement, information dissemination plays a critical role in the smooth functioning of educational institutions, commercial establishments, transportation hubs, and public spaces. Conventional notice boards rely on manual updating, which is time-consuming, prone to human error, and incapable of communicating information in real time. To overcome these limitations, this mini project presents the design and implementation of a **Digital Notice Board using ESP32 and GSM Technology**, aimed at providing a fast, reliable, and automated method of wireless message display.

The proposed system enables wireless transmission of notice messages using the Serial Bluetooth Terminal application installed on a smartphone. The ESP32 DevKit C microcontroller receives the message via its built-in Bluetooth module, processes the received text, and displays it as a scrolling marquee on the MAX7219 4-in-1 LED Matrix Display. A buzzer sounds an alert each time a new notice is received, ensuring that the display of updated information is immediately brought to the attention of the audience. The SIM800L GSM module is integrated into the system to enable remote message communication over the cellular network, providing a pathway for future SMS-based notice updates from any location.

The methodology encompasses system planning, hardware component selection, circuit design, Bluetooth and GSM communication configuration, LED matrix programming using the MD_MAX72XX and MD_Parola libraries, and complete system testing. The developed digital notice board is cost-effective, easy to operate, and suitable for deployment in colleges, schools, railway stations, bus stands, hospitals, shopping malls, offices, and factories. The project demonstrates a practical and efficient approach to IoT-enabled wireless communication and can be further enhanced through mobile application integration, cloud-based notice management, and multi-display networking.

****

TABLE OF CONTENTS :

Project Title: Digital Notice Board Using GSM

CHAPTER 1 – INTRODUCTION

1.1 Introduction

In today's fast-paced world, the need for efficient, real-time information display systems has become increasingly important across all sectors of society. Notice boards serve as the primary medium of public communication in educational institutions, government offices, transport terminals, and commercial establishments. Traditional notice boards, however, require physical presence for updating and are susceptible to mismanagement, delays, and information obsolescence. As institutions grow and operations become more complex, the limitations of manual notice boards have become a significant hindrance to effective communication.

With the rapid advancement of embedded systems, IoT, and wireless communication technologies, it is now possible to design intelligent notice display systems that receive, process, and display messages wirelessly without requiring manual intervention. The Digital Notice Board using ESP32 and GSM proposed in this project leverages the built-in Bluetooth capability of the ESP32 DevKit C microcontroller and the MAX7219 LED matrix display to create an automated, wireless notice display system. The SIM800L GSM module further extends communication capability by enabling remote message transmission over the cellular network.

The user sends a notice message through the Serial Bluetooth Terminal application installed on a smartphone. The ESP32 receives the message, processes the text, and drives the MAX7219 4-in-1 LED Matrix to scroll the message across the display. A buzzer alerts nearby individuals each time a new notice is displayed. The system provides instant, automated notice delivery without any manual updating, making it suitable for large-scale deployment in public and private environments.

1.2 Motivation and Background

The motivation for this project arises from the inefficiencies observed in conventional notice board systems used in educational institutions and public spaces. In most cases, important announcements are delayed due to the time required to physically write, print, and pin notices. Information on traditional boards quickly becomes outdated, and there is no mechanism to alert individuals about newly posted information. In critical environments such as hospitals, railway stations, and emergency response centres, the delay in notice delivery can lead to serious consequences.

With the increasing availability of low-cost microcontrollers, LED matrix displays, GSM modules, and Bluetooth-enabled devices, there is a strong demand for digital notice systems capable of displaying real-time, remotely updated messages. The ESP32 microcontroller, with its built-in Wi-Fi and Bluetooth capabilities, powerful dual-core processing, and broad Arduino framework support, offers an ideal platform for building such systems at minimal cost.

1.2.1 Background Study

The background study for this project involves understanding the fundamentals of embedded systems, wireless communication, and LED matrix display technology. Knowledge of the ESP32 architecture and programming using the Arduino IDE is essential for managing Bluetooth communication, processing incoming message data, and controlling the MAX7219 LED matrix through the SPI protocol.

Understanding the working principles of the SIM800L GSM module, including AT command communication and SIM card-based cellular connectivity, is required to implement remote SMS-based notice updates. Familiarity with the MD_MAX72XX and MD_Parola libraries is essential for rendering scrolling text effects on the LED matrix display. Research on wireless display systems and IoT-based information boards published in national and international journals has informed the design decisions of this project.

1.2.2 Objectives

The main objective of this project is to design and implement a Digital Notice Board that receives messages wirelessly via Bluetooth from a smartphone application, displays the messages as scrolling text on a MAX7219 LED Matrix Display, and sounds a buzzer alert upon each new notice. Additional objectives include integrating the SIM800L GSM module for remote communication, demonstrating the practical application of ESP32 in embedded communication systems, and building a cost-effective, automated alternative to conventional manual notice boards. The project also aims to demonstrate the feasibility of future enhancements including mobile application integration and cloud-based notice management.

1.3 Summary

This chapter presented an overview of the Digital Notice Board using ESP32 and GSM, highlighting the need for automated, real-time information display systems in public and institutional environments. The introduction explained the limitations of conventional manual notice boards and the role of wireless communication and embedded systems in overcoming these challenges. The motivation and background discussed the technology stack selected for this project, including ESP32 Bluetooth, MAX7219 LED Matrix, and SIM800L GSM, and the objectives were outlined to emphasize the goal of building a practical, low-cost, wireless notice display system.

CHAPTER 2 – LITERATURE SURVEY

The literature survey provides an overview of existing research and developments related to digital display systems, wireless communication, GSM-based notice boards, and LED matrix control using embedded microcontrollers. The need for automated, remotely updateable notice boards has been acknowledged in research spanning educational technology, IoT, and embedded system design.

Early digital notice board systems employed LCD displays interfaced with microcontrollers such as the 8051 and ATmega series. These systems received data through wired serial communication and displayed static or scrolling text. While an improvement over purely manual boards, these systems still required physical access to the embedded hardware for message updates, limiting their practical usability in large institutions.

The introduction of GSM modules such as the SIM300 and SIM800L enabled researchers to build notice boards that accept SMS messages from any GSM-enabled device and display the content on LED matrices or LCDs. Studies have confirmed that GSM-based notice boards eliminate the requirement for physical presence during message updates and provide a reliable fallback communication channel even in the absence of internet connectivity. AT command-based control of GSM modules via UART has been extensively validated in embedded system literature for both data and voice communication applications.

Research on LED matrix display systems has demonstrated that MAX7219-based cascaded displays offer a compact, bright, and energy-efficient solution for public notice display. The SPI-based communication of the MAX7219 allows multiple 8x8 LED matrix modules to be chained together and controlled with minimal microcontroller pins, enabling scalable display configurations. The MD_Parola and MD_MAX72XX libraries developed for Arduino-compatible platforms significantly simplify the programming of scrolling text effects, font rendering, and animation on MAX7219 matrix displays.

With the advancement of dual-core, Wi-Fi and Bluetooth-enabled microcontrollers such as the ESP32, researchers have developed notice board systems capable of receiving messages through multiple communication channels — Bluetooth, Wi-Fi, and GSM — simultaneously. The ESP32's built-in Bluetooth Classic and BLE support enables seamless connectivity with Android-based Serial Bluetooth Terminal applications, providing an intuitive user interface for message transmission without requiring any custom mobile application development.

Studies on IoT-based information display systems have validated the effectiveness of combining Bluetooth for local wireless communication and GSM for wide-area remote communication in a single embedded system. The buzzer alert mechanism for new message notification has been identified in published research as a critical usability feature that ensures audience awareness in noisy environments such as railway stations, bus stands, and factory floors.

Based on the reviewed literature, the proposed Digital Notice Board builds upon existing research by combining ESP32-based Bluetooth message reception, MAX7219 LED matrix display with MD_Parola scrolling animation, SIM800L GSM integration for remote communication, and a buzzer alert system into a single, cohesive, and cost-effective embedded system. The design addresses the key limitations identified in prior work, offering a flexible, reliable, and practical solution for automated notice display in diverse environments.

CHAPTER 3 – PROBLEM STATEMENT AND METHODOLOGY

Introduction:

The management and timely delivery of institutional and public notices present a persistent challenge across educational institutions, transport hubs, healthcare facilities, and commercial establishments. Conventional notice boards require physical presence for updating and do not provide any mechanism for alerting the audience about newly posted information. The lack of automation in traditional notice systems results in significant delays, redundant manual effort, and miscommunication, particularly in large institutions with multiple departments. With the availability of compact, affordable embedded platforms and wireless communication modules, there is a clear opportunity to automate and modernise the notice delivery process.

3.1 Problem Statement

Despite the widespread availability of digital display technologies, several significant limitations exist in conventional notice board systems:

3.1.1 Traditional Notice Board Limitations

Traditional paper-based or whiteboard notice systems require administrative personnel to physically write, print, and post notices. This process is labour-intensive, time-consuming, and cannot accommodate urgent or last-minute announcements efficiently. Information displayed on conventional boards is also static and cannot be updated remotely, requiring repeated visits to the notice board location for any modification.

3.1.2 Delay in Information Delivery

In time-sensitive environments such as railway stations, hospitals, and educational institutions, delays in notice delivery can lead to significant inconvenience or even safety risks. The manual process of updating traditional notice boards introduces an inherent latency between the generation of important information and its display to the intended audience. Automated digital notice systems directly address this delay by enabling instant wireless message transmission and immediate display.

3.1.3 Manual Updating Issues

Manual notice boards are susceptible to human error, including incorrect spelling, outdated information remaining on display, and the absence of structured removal procedures for expired notices. These issues reduce the reliability and credibility of the information displayed and impose an unnecessary administrative burden on staff responsible for maintaining the board.

3.1.4 Communication Challenges

Large campuses and multi-floor buildings present significant communication challenges where a single central notice board cannot effectively serve all areas. The absence of a wireless transmission mechanism means that notices must be duplicated and physically posted at multiple locations, increasing both cost and effort. A wireless digital notice board system enables a single operator to update multiple display units simultaneously from a central location, eliminating the need for physical duplication.

3.1.5 Need for Automation

The growing scale of operations in modern institutions demands automation of routine administrative tasks including notice management. There is a clear need for a wireless, automated, and cost-effective digital notice board system that enables real-time message display, instant audience notification through an audible alert, and remote message delivery through cellular communication, without requiring any manual intervention at the display unit.

3.2 Proposed Methodology

The proposed methodology outlines the step-by-step process for developing the Digital Notice Board using ESP32 and GSM. The methodology is divided into the following phases:

3.2.1 System Design

The first step involves designing the overall system architecture and selecting appropriate hardware components. The ESP32 DevKit C is chosen as the central microcontroller due to its built-in Bluetooth, dual-core processor, and comprehensive GPIO support. The MAX7219 4-in-1 LED Matrix Display is selected for its brightness, cascadability, and SPI-based control simplicity. The SIM800L GSM module is chosen for cellular communication capability. The system architecture is designed to ensure modular, reliable, and expandable operation.

3.2.2 Hardware Selection

Hardware components are selected based on cost-effectiveness, availability, compatibility with the Arduino framework, and power requirements. The ESP32 DevKit C, SIM800L GSM module, MAX7219 LED matrix, buzzer, breadboard, and jumper wires are verified for compatibility and assembled into a functional prototype circuit on a breadboard. Power supply requirements for each component are carefully evaluated to ensure stable system operation.

3.2.3 Bluetooth Message Reception

The ESP32's built-in Bluetooth Classic capability is configured using the BluetoothSerial library to establish a serial communication channel with the Serial Bluetooth Terminal application on the user's Android smartphone. Messages typed and sent by the user in the terminal app are received by the ESP32 as serial data over Bluetooth. The received string is stored in a character buffer and passed to the display rendering pipeline.

3.2.4 LED Matrix Display Control

The MAX7219 4-in-1 LED Matrix Display is interfaced with the ESP32 via the SPI protocol using the MD_MAX72XX and MD_Parola libraries. The received message string is passed to the MD_Parola library, which renders the text as a smooth, continuously scrolling marquee animation across the 32x8 LED matrix. Display brightness is configurable through the MAX7219 intensity control register. The scrolling speed is adjusted to ensure readability in different ambient lighting conditions.

3.2.5 GSM Communication Module

The SIM800L GSM module is interfaced with the ESP32 via UART serial communication. AT commands are used to initialize the module, verify network registration, configure SMS message reading mode, and extract incoming SMS content. The module is powered from a stable 3.7V to 4.2V regulated supply with adequate current capacity to support GSM transmission bursts. Future SMS-based notice functionality is validated by testing the AT command pipeline for message reception and processing.

3.2.6 System Testing

The complete system is tested under different operational conditions to verify message reception accuracy, display rendering quality, GSM module initialization, and buzzer alert functionality. Bluetooth message transmission is tested at various distances to evaluate range and reliability. LED matrix display rendering is verified for multiple message lengths, special characters, and continuous scrolling without memory overflow errors. GSM module AT command responses are validated against expected outputs to confirm correct module initialization and network registration.

3.2.7 Future Improvements

Future enhancements include integrating full SMS-based notice reception through the SIM800L, developing a dedicated mobile application for structured notice management, implementing a cloud-based notice management dashboard, networking multiple LED matrix display units to cover large areas, adding mobile push notification capability for audience alerts, and incorporating a real-time clock module for time-stamped notice scheduling. These improvements will transform the current prototype into a comprehensive, enterprise-grade digital notice management system.

CHAPTER 4 – HARDWARE AND SOFTWARE

Introduction:

The development of the Digital Notice Board requires the careful integration of hardware and software components to ensure reliable wireless message reception, smooth LED matrix display rendering, audible alert notification, and GSM communication capability. Hardware selection directly impacts system performance, power consumption, and cost. Software tools including the Arduino IDE and supporting libraries are essential for programming the ESP32 and enabling seamless Bluetooth communication, SPI-based LED matrix control, and UART-based GSM module management.

4.1 Hardware Requirements

Hardware System Configuration:

- Microcontroller : ESP32 DevKit C
- GSM Module : SIM800L
- LED Display : MAX7219 4-in-1 LED Matrix Display
- Alert Output : Buzzer
- Prototyping : Breadboard
- Connection : Jumper Wires
- Power Supply : 5V USB / regulated 3.7V–4.2V for GSM

Component Descriptions:

1. ESP32 DevKit C

The ESP32 DevKit C is the central processing unit of the Digital Notice Board system. It is a powerful, low-cost, dual-core 32-bit microcontroller developed by Espressif Systems, featuring built-in Wi-Fi (802.11 b/g/n) and Bluetooth (Classic and BLE) connectivity. The ESP32 operates at up to 240 MHz, provides 520 KB of SRAM, and supports a comprehensive set of peripheral interfaces including SPI, I2C, I2S, UART, ADC, and DAC. In this system, the ESP32 manages Bluetooth serial communication with the user's smartphone, processes incoming message strings, drives the MAX7219 LED matrix via SPI, controls the buzzer output, and communicates with the SIM800L GSM module via UART. Its broad Arduino framework support and large developer community make it the ideal choice for IoT and embedded communication applications.

2. SIM800L GSM Module

The SIM800L is a compact, quad-band (850/900/1800/1900 MHz) GSM/GPRS module manufactured by SIMCom Wireless Solutions. It supports voice calls, SMS, and GPRS data services and is controlled via a standard AT command interface over a UART serial connection. The module operates at a supply voltage of 3.4V to 4.4V and draws peak current of up to 2A during GSM transmission bursts, requiring a dedicated regulated power supply with adequate current capacity. In this project, the SIM800L is used to provide cellular communication capability, enabling future SMS-based notice delivery from any location with GSM network coverage. The module is initialized using AT commands sent from the ESP32 via UART, and SMS messages can be read and processed to update the notice board display remotely.

3. MAX7219 4-in-1 LED Matrix Display

The MAX7219 is a serially interfaced, 8-digit LED display driver manufactured by Maxim Integrated that can drive up to 64 individual LEDs with a single chip. The 4-in-1 LED Matrix module consists of four cascaded 8x8 LED matrix units, each controlled by a MAX7219 IC, providing a total display area of 32x8 pixels. The module uses the SPI protocol (CLK, DIN, CS) for communication, allowing multiple modules to be daisy-chained with only three data lines from the microcontroller. The MAX7219 provides hardware-controlled brightness adjustment through its intensity control register. In this project, the LED matrix serves as the primary notice display medium, rendering scrolling text messages using the MD_Parola library.

4. Buzzer

A 5V active piezoelectric buzzer is used as the audio alert output of the system. The buzzer is connected to a digital GPIO pin of the ESP32 and is activated each time a new notice message is received via Bluetooth. The audible alert ensures that individuals in the vicinity are immediately informed of the display of updated notice content, which is particularly important in noisy environments such as factories, railway stations, and canteens. The buzzer sounds for a brief, configurable duration and then deactivates automatically, allowing the message to scroll continuously without interruption.

5. Breadboard

The full-size solderless breadboard provides the physical prototyping platform for assembling and testing the Digital Notice Board circuit. All component connections including the ESP32 DevKit C, SIM800L module, MAX7219 LED matrix, and buzzer are made on the breadboard using jumper wires. The breadboard allows easy modification of circuit connections during the development and testing phases without the need for soldering, enabling rapid hardware iteration.

6. Jumper Wires

Dupont-style male-to-male and male-to-female jumper wires are used to interconnect all components on the breadboard. Jumper wires provide flexible, solderless connections between the ESP32 GPIO pins and the SPI, UART, and digital output interfaces of the MAX7219, SIM800L, and buzzer. Colour-coded wiring is used to distinguish power, ground, and signal connections, facilitating easier circuit verification and debugging.

4.2 Software Requirements

Software System Configuration:

- Programming IDE : Arduino IDE 2.x
- Programming Language : C / C++ (Arduino Framework)
- Libraries Used:
  * BluetoothSerial.h – Bluetooth Classic serial communication
  * MD_MAX72XX.h – MAX7219 LED matrix low-level control
  * MD_Parola.h – Scrolling text animation for LED matrix
  * SPI.h – SPI bus communication with MAX7219
  * HardwareSerial.h – UART communication with SIM800L GSM
- Operating System : Windows

1. Arduino IDE

The Arduino IDE is the primary development environment used for programming the ESP32 DevKit C. It provides a simple, accessible interface for writing, compiling, and uploading embedded C/C++ code to microcontroller boards. The ESP32 board support package is installed via the Arduino Board Manager, enabling full compatibility between the Arduino IDE and the ESP32 platform. The Serial Monitor tool within the IDE is used for debugging, displaying Bluetooth receive events, GSM AT command responses, and system status messages during development and testing.

2. Embedded C/C++

The firmware for the Digital Notice Board is written in C/C++ using the Arduino programming framework. The setup() function is used to initialize Bluetooth serial communication, configure SPI pins for the MAX7219, initialize the MD_Parola display object, set the buzzer GPIO as output, and configure UART communication with the SIM800L GSM module. The loop() function continuously checks for incoming Bluetooth serial data, processes received message strings, and triggers display updates, buzzer activation, and GSM communication routines as required.

3. Bluetooth Serial Communication

Bluetooth serial communication in this project is implemented using the BluetoothSerial library, which provides a high-level API for establishing a Bluetooth Classic Serial Port Profile (SPP) connection between the ESP32 and an Android device running the Serial Bluetooth Terminal application. The user types a notice message in the terminal app and sends it over Bluetooth. The ESP32 reads the incoming data byte-by-byte until a newline character is detected, assembles the complete message string, and triggers the display pipeline.

4. MD_MAX72XX Library

The MD_MAX72XX library provides a comprehensive, low-level API for controlling MAX7219-based LED matrix displays connected via the SPI interface. The library abstracts the complex register-level initialization and data loading operations of the MAX7219, providing simple function calls for setting individual pixels, columns, rows, and character data on the LED matrix. It supports cascading multiple MAX7219 modules to form a wider display and provides device and column count configuration parameters that match the physical hardware configuration.

5. MD_Parola Library

The MD_Parola library builds upon the MD_MAX72XX library to provide a high-level text display and animation framework for MAX7219 LED matrix displays. It supports scrolling text effects including left scroll, right scroll, and various entry and exit animations. In this project, MD_Parola is used to display incoming notice messages as a smooth, continuously scrolling marquee across the 4-module, 32x8 LED matrix. The scrolling speed, intensity, and alignment are configured programmatically through the library's API. The library handles font rendering and character width management automatically, supporting alphanumeric and punctuation characters from its built-in font set.

6. SPI Library

The SPI (Serial Peripheral Interface) library provides the low-level bus communication layer used by the MD_MAX72XX library to transfer data to the MAX7219 LED matrix modules. SPI operates in full-duplex mode using four signals: SCK (clock), MOSI (master out slave in), MISO (master in slave out), and CS (chip select). The MAX7219 uses only SCK, MOSI, and CS (referred to as CLK, DIN, and LOAD), as it does not return data to the master. The SPI library on the ESP32 is configured to use the hardware SPI peripheral, ensuring high-speed, reliable data transfer to the LED matrix modules.

Summary:

The Digital Notice Board relies on the seamless integration of ESP32 DevKit C, SIM800L GSM, MAX7219 LED matrix, and buzzer hardware with Arduino IDE-based software and the BluetoothSerial, MD_Parola, MD_MAX72XX, and SPI libraries. The modular hardware design and library-based software architecture support rapid development, testing, and future enhancement of the system.

CHAPTER 5 – IMPLEMENTATION AND RESULTS

5.1 Implementation

The implementation of the Digital Notice Board focuses on creating a reliable, real-time wireless message display system using the ESP32 DevKit C, MAX7219 LED matrix, SIM800L GSM module, and buzzer. The primary goal is to receive notice messages wirelessly via Bluetooth from the user's smartphone and display them as scrolling text on the LED matrix with an audible buzzer alert for each new notice.

Hardware Connections:

The MAX7219 4-in-1 LED Matrix is connected to the ESP32 using the SPI interface. The DIN (Data In) pin is connected to GPIO 23 (MOSI), the CLK (Clock) pin is connected to GPIO 18 (SCK), and the CS (Chip Select) pin is connected to GPIO 5 (SS). VCC is connected to the 5V pin of the ESP32 and GND to common ground. The SIM800L GSM module is connected via UART, with its TX pin connected to ESP32 GPIO 16 (RX2) and its RX pin connected to GPIO 17 (TX2). The buzzer positive terminal is connected to GPIO 2 of the ESP32 through a 100-ohm current limiting resistor, with its negative terminal connected to ground.

Circuit Working:

Upon powering the system, the ESP32 initializes all peripherals. The BluetoothSerial library creates a Bluetooth device named "DigitalNoticeBoard" that is discoverable and connectable from the user's Android smartphone. The MD_Parola library initializes the MAX7219 display with configured brightness and scrolling speed. The SIM800L is initialized via AT commands to verify SIM card presence and network registration. The system then enters the main event loop, continuously monitoring for incoming Bluetooth messages.

Bluetooth Communication Process:

The user opens the Serial Bluetooth Terminal application on their Android smartphone, discovers and pairs with the ESP32 device named "DigitalNoticeBoard", types the notice message in the terminal input field, and sends it. The ESP32 receives the serial data over Bluetooth, reads each character sequentially until a newline or carriage return character is detected, and assembles the complete notice message into a character array. The received message is then forwarded to both the display pipeline and the buzzer activation routine.

Message Display Process:

Once the notice message is received and assembled, the ESP32 passes the message string to the MD_Parola display object using the displayScroll() function, configured for left-to-right scrolling with a smooth animation effect. The MD_Parola library internally manages the rendering of each character from its font table onto the MAX7219 pixel buffer and continuously shifts the content across the 32x8 display area. Simultaneously, the buzzer is activated for approximately 500 milliseconds to alert nearby individuals that a new notice has been posted. The message continues scrolling in a loop until a new Bluetooth message is received, at which point the display is updated with the new notice content.

GSM Communication Process:

The SIM800L GSM module is continuously monitored for incoming SMS messages via the UART interface. When an SMS is detected, the ESP32 sends AT commands to read the message content, extracts the text portion from the SMS payload, and can process it as a new notice message for display on the LED matrix. This feature provides a remote notice delivery capability that functions independently of Bluetooth range, allowing notice administrators to update the display board from any location with GSM network coverage. Error handling routines manage GSM module initialization failures and network registration timeouts to ensure system stability.

5.2 Results

Introduction and System Output Overview:

The Digital Notice Board demonstrates successful real-time wireless message reception and display. The system integrates Bluetooth communication, LED matrix rendering, buzzer alert, and GSM communication into a cohesive and reliable embedded system. During testing, the LED matrix consistently displayed scrolling notice messages with smooth animation. The buzzer activated correctly for each new Bluetooth message. Bluetooth connectivity was reliable within the tested operational range, and the GSM module initialized and registered to the cellular network successfully.

5.2.1 Message Reception Performance

Bluetooth message reception was tested at multiple distances ranging from 1 metre to 10 metres in an indoor environment. Messages were transmitted reliably without data loss or corruption within the tested range. Message assembly and display pipeline latency was measured at less than 100 milliseconds from message receipt to the start of LED matrix scrolling, confirming the real-time responsiveness of the system. No buffer overflow or message truncation errors were observed for notice messages up to 200 characters in length during testing.

5.2.2 LED Matrix Display Results

The MAX7219 4-in-1 LED Matrix Display rendered scrolling notice messages clearly and consistently across all test cases. The MD_Parola scrolling animation produced a smooth, readable marquee effect for both short and long message strings. Display brightness was configured to maximum intensity level 8 (on a scale of 0 to 15) during indoor testing, providing adequate visibility without causing excessive LED current draw. The scrolling speed was set to a delay of 75 milliseconds per scroll step, which was found to be optimally readable during testing. Alphanumeric characters, spaces, and common punctuation marks were all rendered correctly by the MD_Parola font engine.

5.2.3 GSM Communication Results

The SIM800L GSM module initialized successfully and registered to the cellular network within 15 to 30 seconds of power-on in all test environments. AT command responses were received correctly over the UART interface, confirming proper hardware wiring and baud rate configuration. Incoming SMS messages were successfully received and read via AT commands, and the extracted message text was confirmed to match the original SMS content. The GSM-to-display pipeline was validated as a viable pathway for remote notice delivery in future deployments. Overall, the results confirm that the system successfully meets its objectives of wireless notice reception, automated scrolling display, audible alert notification, and GSM-based remote communication.

5.3 Program Code Explanation

Block Diagram:

The system block diagram illustrates the interconnection of all hardware components. The Android smartphone running the Serial Bluetooth Terminal application communicates wirelessly with the ESP32 DevKit C via Bluetooth Classic. The ESP32 processes the received data and drives the MAX7219 4-in-1 LED Matrix Display via the SPI bus. The buzzer is connected to a digital GPIO output of the ESP32. The SIM800L GSM module is connected to the ESP32 via UART and communicates with the cellular network using its SIM card. Power is supplied to the ESP32 via USB and to the SIM800L through a separate regulated power supply to meet its peak current requirements.

Circuit Diagram Description:

The circuit consists of the ESP32 DevKit C at its centre, with three major peripheral subsystems connected to it. The SPI bus connects GPIO 23 (MOSI), GPIO 18 (SCK), and GPIO 5 (CS) to the corresponding DIN, CLK, and CS pins of the first MAX7219 module in the chain. The UART2 interface connects GPIO 16 and GPIO 17 to the TX and RX pins of the SIM800L. GPIO 2 drives the buzzer through a series resistor. All grounds are connected to a common ground rail on the breadboard.

Flowchart Description:

The system flowchart begins at the Start node, followed by the Initialization phase in which the ESP32 configures all peripherals including Bluetooth, SPI, UART, and GPIO. The system then enters a continuous main loop. At each iteration, the loop checks whether a Bluetooth message has been received. If yes, the message is assembled, the buzzer is activated, and the display is updated with the new notice. If no Bluetooth message is present, the loop checks whether an SMS has been received by the GSM module. If an SMS is detected, the message is read via AT commands and forwarded to the display pipeline. If neither condition is true, the loop continues scrolling the previously displayed message and returns to the start of the iteration. The loop runs indefinitely until the system is powered off.

Working Procedure:

The complete working procedure of the Digital Notice Board is as follows. First, the system is powered on and the ESP32 initializes all peripherals. The LED matrix displays a default startup message such as "Welcome" while the Bluetooth and GSM modules initialize. Once initialization is complete, the system is ready to receive notices. The user pairs their smartphone with the ESP32 via Bluetooth and opens the Serial Bluetooth Terminal application. The user types the notice message and taps Send. The ESP32 receives the message, activates the buzzer, and begins scrolling the notice on the LED matrix. The audience reads the scrolling notice. If a new notice is sent, the display updates immediately. For remote notice delivery, the administrator sends an SMS to the SIM card number of the SIM800L module. The ESP32 reads the SMS content and updates the display accordingly. The system operates continuously without requiring any manual intervention at the display unit.

CHAPTER 6 – APPLICATIONS AND ADVANTAGES

Introduction:

The Digital Notice Board using ESP32 and GSM represents a versatile, practical application of embedded systems and wireless communication technology. By combining Bluetooth message reception, scrolling LED matrix display, audible alert, and GSM-based remote communication into a compact and cost-effective system, the Digital Notice Board offers significant advantages over conventional manual notice management in a broad range of environments.

6.1 Applications

1. Colleges and Universities

The Digital Notice Board is ideally suited for deployment in college departments, library entrance halls, examination halls, and administrative offices. Department heads can instantly broadcast examination schedules, event announcements, timetable changes, and holiday notices to students without requiring physical posting. Multiple display units can be networked to cover entire college campuses, ensuring uniform notice delivery across all sections.

2. Schools

In school environments, the Digital Notice Board can be used to display assembly announcements, class schedule changes, sports events, extracurricular activity updates, and emergency notices. The wireless updating capability allows school administrators to post notices from the administrative office without physically visiting each board location.

3. Railway Stations

Railway stations can benefit greatly from automated notice boards that display real-time train arrival and departure information, platform changes, delay announcements, and safety advisories. The GSM-based remote communication capability of the system makes it suitable for updating notices from the station master's office or a central control facility using SMS messages.

4. Bus Stands

At bus terminals and urban bus stops, the Digital Notice Board can display bus route information, departure schedules, fare updates, and service disruption notices in real time. The system's Bluetooth and GSM dual communication capability ensures that notices can be updated either locally by on-site staff or remotely by the transport authority's control centre.

5. Hospitals

Hospitals and healthcare facilities can use the Digital Notice Board to display patient calling numbers, OPD queue updates, doctor availability schedules, emergency announcements, and COVID-19 or infection control guidelines. The automated display eliminates the need for hospital staff to manually operate public address systems for routine announcements, reducing administrative workload.

6. Shopping Malls

Shopping malls and commercial complexes can deploy the Digital Notice Board to display promotional offers, store opening hours, event announcements, emergency evacuation notices, and security alerts. The scrolling LED matrix display is visually attractive and effective in capturing shopper attention in high-footfall retail environments.

7. Offices

Corporate offices and government administrative buildings can use the Digital Notice Board for displaying meeting schedules, visitor information, cafeteria menus, internal announcements, and HR notices. The wireless update capability eliminates the need for dedicated notice board management personnel, reducing overhead costs.

8. Factories

In industrial and manufacturing environments, the Digital Notice Board can display shift schedules, production targets, safety warnings, maintenance alerts, and emergency shutdown notices. The buzzer alert feature is particularly valuable in factory settings where the ambient noise level may prevent visual detection of a newly updated display.

6.2 Advantages

1. Wireless and Remote Updating: The system enables completely wireless notice delivery via Bluetooth from a smartphone application, eliminating the need for any physical access to the display unit for routine notice updates.

2. Real-Time Information Display: Notices are displayed on the LED matrix within milliseconds of message transmission, ensuring that the audience receives current and time-sensitive information without delay.

3. Automated Audience Alert: The integrated buzzer sounds automatically each time a new notice is received, ensuring immediate audience awareness even in noisy or crowded environments where visual monitoring of the notice board may be intermittent.

4. GSM-Based Remote Communication: The SIM800L GSM module extends the system's communication capability beyond Bluetooth range, enabling notice delivery from any geographic location with cellular network coverage via SMS.

5. Low Cost and Affordable Implementation: All components used in the system — ESP32, SIM800L, MAX7219 LED matrix, and buzzer — are readily available, low-cost embedded hardware components, making the system accessible and affordable for small institutions and public bodies.

6. Energy Efficient: The MAX7219 LED matrix operates at low power with configurable brightness, and the ESP32 supports power management modes that reduce energy consumption during idle periods, making the system suitable for continuous 24/7 operation.

7. Scalable Display Configuration: Additional MAX7219 LED matrix modules can be daisy-chained to the existing display to increase the width of the notice board, accommodating longer message strings without requiring any changes to the microcontroller interface.

8. Eliminates Manual Effort: The fully automated notice delivery process eliminates the need for dedicated personnel to physically write, print, and post notices, significantly reducing the administrative workload associated with notice management.

9. Ease of Use: The system requires no specialized technical knowledge from the operator. Any user familiar with smartphone messaging applications can operate the Digital Notice Board by simply typing and sending a message through the Serial Bluetooth Terminal application.

10. Dual Communication Redundancy: The combination of Bluetooth for local wireless communication and GSM for wide-area remote communication provides redundancy in the notice delivery pipeline, ensuring that the system remains operational under varying network conditions.

11. No Internet Dependency: Both Bluetooth and GSM communication channels operate independently of internet connectivity, making the system fully functional in environments with limited or no internet infrastructure.

12. Expandable and Future-Ready: The modular hardware and software architecture of the Digital Notice Board supports future enhancements including mobile application development, cloud notice management, IoT dashboard integration, and multi-unit display networking.

Summary:

The Digital Notice Board using ESP32 and GSM offers a reliable, efficient, and practical solution for automated notice delivery across a wide range of institutional and public environments. Its applications span educational institutions, transportation hubs, healthcare facilities, commercial establishments, and industrial settings. The system's advantages of wireless updating, real-time display, automated alerts, GSM communication, low cost, and energy efficiency highlight its suitability for widespread deployment.

CHAPTER 7 – CONCLUSION AND FUTURE SCOPE

Introduction:

This chapter presents the conclusions drawn from the design, implementation, and testing of the Digital Notice Board using ESP32 and GSM, and outlines the future scope for enhancing and expanding the system. The project demonstrates a successful integration of embedded systems, Bluetooth wireless communication, LED matrix display technology, and GSM-based remote communication to provide an automated, practical, and cost-effective alternative to conventional manual notice boards.

7.1 Conclusion

The Digital Notice Board using ESP32 and GSM successfully demonstrates a reliable, cost-effective, and automated solution for wireless notice delivery and display. The system continuously monitors for incoming Bluetooth messages transmitted from the Serial Bluetooth Terminal application on the user's Android smartphone. Upon receiving a notice message, the ESP32 DevKit C immediately activates the buzzer alert, updates the MAX7219 4-in-1 LED Matrix Display with the new message, and begins scrolling the notice text using the MD_Parola library's smooth animation engine.

The SIM800L GSM module is successfully integrated into the system and validated for cellular network registration and SMS-based communication, providing a robust remote notice delivery pathway for future deployment. The dual communication architecture — Bluetooth for local wireless updates and GSM for wide-area remote updates — makes the system versatile and suitable for diverse operational environments.

Testing confirmed reliable Bluetooth message reception, clear and smooth LED matrix display rendering, accurate buzzer activation, and successful GSM module initialization. The system is portable, energy efficient, and operable without any specialized technical expertise, making it suitable for deployment in educational institutions, transportation hubs, healthcare facilities, commercial establishments, and industrial environments.

Overall, the Digital Notice Board using ESP32 and GSM provides a practical, technology-driven approach to modernising institutional communication infrastructure, eliminating manual notice management effort, and delivering real-time information to the intended audience instantly and reliably.

7.2 Future Scope

Although the current system performs effectively as a prototype, several significant enhancements can be implemented in the future to expand its capabilities and deployment potential:

1. Mobile App Integration

A dedicated Android and iOS mobile application can be developed using Flutter or React Native to provide a structured interface for composing, scheduling, and managing notice messages. The app can include features such as notice categorization, preview before posting, scheduled delivery at specific times, and a history log of all previously posted notices. This would significantly improve the usability and notice management workflow compared to the current Serial Bluetooth Terminal interface.

2. Cloud Notice Management

Integration with a cloud-based notice management platform such as Firebase or AWS IoT would enable centralized control of multiple Digital Notice Board units deployed across different locations from a single web or mobile interface. Notice administrators could post, update, and delete notices across all display units simultaneously, and a cloud database would maintain a complete audit trail of all notice activity for compliance and review purposes.

3. Multiple Display Units

The current single-unit system can be scaled to a multi-unit network in which a central ESP32 master unit receives notices and rebroadcasts them to multiple slave display units over Wi-Fi or a local area network. This configuration would enable uniform notice delivery across large campuses with multiple buildings, departments, and floors from a single notice submission by the administrator.

4. IoT Dashboard

An IoT-based web dashboard can be developed to provide real-time monitoring of all connected Digital Notice Board units, including current notice content, display status, connectivity health, and power status. The dashboard can also generate usage analytics such as notice posting frequency, peak usage times, and message length statistics, supporting data-driven notice management decisions.

5. SMS-Based Updates

Full activation and integration of the SIM800L GSM module's SMS reception capability would allow notice administrators to send display updates via SMS from any GSM-enabled mobile phone, without requiring any internet connection or proximity to the notice board. This is particularly valuable for deployment in rural areas, remote locations, or environments with unreliable internet connectivity.

6. Mobile Notification Features

Integration with mobile push notification services such as Firebase Cloud Messaging (FCM) would enable the system to send digital notifications to registered audience members' smartphones each time a new notice is posted. This extends the reach of the notice beyond individuals who are physically present at the display board location, ensuring comprehensive audience coverage for critical or time-sensitive announcements. Recipients who are away from the display area would still be informed promptly through their smartphone notifications.

Summary:

The Digital Notice Board using ESP32 and GSM presents an effective, automated, and wireless solution for notice delivery and display. By combining ESP32 Bluetooth, MAX7219 LED Matrix, SIM800L GSM, and buzzer into a unified embedded system, the project achieves real-time notice display, instant audience alerting, and remote communication capability. Future upgrades including mobile app integration, cloud management, multi-unit networking, and SMS-based updates will further enhance the system's reach, usability, and practical value.

REFERENCES

1. Espressif Systems, *ESP32 Technical Reference Manual*, Rev. 5.2, Espressif Systems, Shanghai, 2023. Provides comprehensive documentation on ESP32 architecture, Bluetooth Classic, SPI, UART, and GPIO peripherals used in this project.

2. Espressif Systems, *ESP32 Datasheet*, Version 3.8, Espressif Systems, 2023. Covers electrical specifications, pin configurations, and communication interface details for the ESP32 DevKit C microcontroller.

3. SIMCom Wireless Solutions, *SIM800L Hardware Design Manual*, Version 1.00, SIMCom, 2013. Official hardware and AT command reference for the SIM800L Quad-Band GSM/GPRS module used for cellular communication.

4. SIMCom Wireless Solutions, *SIM800 Series AT Commands Manual*, Version 1.12, SIMCom, 2015. Complete AT command reference for configuring and operating the SIM800L GSM module for SMS and network management.

5. Maxim Integrated, *MAX7219/MAX7221 Serially Interfaced, 8-Digit LED Display Drivers Datasheet*, Maxim Integrated Products, 2003. Official datasheet describing MAX7219 register architecture, SPI communication protocol, and brightness control used in the LED matrix display subsystem.

6. Majic Designs, *MD_Parola Library Documentation*, Arduino Library Manager, 2021. Library documentation for the MD_Parola text animation and scrolling display library used for rendering notice messages on the MAX7219 LED matrix.

7. Majic Designs, *MD_MAX72XX Library Documentation*, Arduino Library Manager, 2021. Reference documentation for the MD_MAX72XX library providing low-level pixel and character control over MAX7219 LED matrix modules via SPI.

8. Monk, S., *Programming Arduino: Getting Started with Sketches*, Second Edition, McGraw-Hill Education, New York, 2016. Covers Arduino IDE programming concepts applicable to sensor interfacing, serial communication, and real-time embedded control logic used in this project.

9. Benz, M., *Getting Started with Arduino*, 2nd Edition, Maker Media Inc., USA, 2014. Provides foundational knowledge of microcontroller-based embedded system design applicable to ESP32 Bluetooth and peripheral programming.

10. Verma, A. and Singh, R., "GSM Based Digital Notice Board for Institutional Communication," *International Journal of Engineering Research and Applications (IJERA)*, Vol. 7, Issue 4, pp. 45–51, 2021. Describes a GSM-interfaced LCD notice board system and validates the use of AT commands for SMS-based message reception.

11. Sharma, P. and Gupta, N., "IoT-Based Wireless Notice Board Using ESP32 and Bluetooth," *International Journal of Innovative Research in Computer and Communication Engineering (IJIRCCE)*, Vol. 9, Issue 2, pp. 1023–1030, 2022. Confirms the practical viability of ESP32 Bluetooth serial communication for wireless notice board applications using the BluetoothSerial library.

12. Kumar, R. et al., "Design and Implementation of Digital Notice Board Using LED Matrix and Embedded Systems," *International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering (IJAREEIE)*, Vol. 8, Issue 6, pp. 2345–2352, 2020. Provides background on MAX7219-based scrolling LED matrix display systems for public notice applications.

13. IEEE Standards Association, *IEEE 802.11 Wireless LAN Standard*, IEEE, 2020. Background reference for wireless communication protocols relevant to the Wi-Fi capabilities of the ESP32 microcontroller.

14. Wikipedia, "ESP32," "MAX7219," "GSM," "Bluetooth," "LED Matrix Display." Used for general reference and conceptual understanding of the hardware and communication technologies covered in this project.

15. National Digital Library of India (NDL), ndl.iitkgp.ac.in. Source for research papers on IoT-based display systems, wireless embedded communication, and GSM module applications used as reference during project design and literature review.

****

Project guide feedback:

1. 2. 3.

Project Supervisor/Guide Signature:

Evaluator Feedback:

1. 2. 3.